

Global Child Hazard Database

The Global Child Hazard Database is a UNICEF-led initiative to provide comprehensive, high-resolution, and globally standardized datasets on children's exposure to multiple hazards. The database aims to improve accessibility and provide actionable insights for country and regional offices, host governments, and other key stakeholders, thereby supporting informed decision-making and policy advocacy.

Core Principles:

The core principles guiding the database's development are:

1. **Relevance:** Data directly relates to the well-being and safety of children.
2. **Availability:** Utilizes data only from open sources and is available for all countries at all administrative levels.
3. **Measurability:** Indicators are quantifiable, spatially distributed and consistent.
4. **Comparability:** Data is standardized to allow for cross-country and regional analysis.
5. **Actionability:** Insights derived from the data can be used to inform and drive protective action.
6. **Child-centric:** The entire framework is built around the unique needs of children

Children's hazard exposure:

The database compiles a wide range of hazard data, primarily focusing on climate and environmental indicators, but also including geophysical hazards and conflict. The data is collected at the highest resolution available to provide a detailed picture of risks at the sub-national level. Key hazards include:

- **Climate Hazards:** Riverine Floods, Coastal Floods, Tropical Storms, Agricultural Drought, Meteorological Drought, Heatwaves, Extreme heat, Fires, Sand and dust storms.
- **Climate-Sensitive Hazards:** Vector-borne diseases (e.g., Malaria), and Air Pollution.
- **Geophysical Hazards:** Earthquakes, Volcanoes, Landslides
- **Complex Emergencies:** Conflict

Using the newly released under18 population layer at high resolution, the database includes estimation of number of children, boys and girls exposed to each one of the above hazards as well as to multiple hazards at administrative levels 0,1 and 2. Exposure analysis shows the potential/probability of children's exposure to a hazard happening in any given year and includes children's population estimates from 2025. The database also includes normalized scores for each hazard at administrative level 2.

Multi-hazard Exposure Analysis:

Two new metrics, multi-hazard count and multi-hazard intensity help us understand the number of children exposed to different number of hazards and at different intensity levels respectively.

Multi-hazard count quantifies the number of distinct climate-sensitive hazards that a specific, smaller geographical area is exposed to (up to a 100-meter grid level, which can be aggregated to different administrative levels). This can be used to quickly pinpoint hotspots where children are exposed to a disproportionately high number of climate hazards.

Multi-hazard intensity can distinguish between areas that experience frequent but mild events and those that face less frequent but more devastating high-intensity events. The indicator goes beyond counting hazards and assesses the severity or magnitude of each hazard, combining them into a composite intensity score at a high resolution.

The database also includes children's exposure to unique hazard combinations such as the total number of children exposed to a specific hazard combination (heatwave, drought and flood or storm and extreme heat) to enable tailored intervention.

Use cases: The standardized database can be utilized for multiple use cases within and outside UNICEF.

- **UNICEF Strategic Plan:** To provide baseline estimates for number of children exposed to multiple hazards and to guide resource allocation for climate, environment and DRR
- **UNICEF Country programme planning:** To support data-driven targeting, sector-specific and hot-spot analysis. It can also be included in CLAC, Sit-An and other country specific needs
- **Government support:** To provide tailored technical support to national governments for policy development, particularly NDCs and NAPs
- **Intergovernmental Collaboration:** To influence key UNFCCC workstreams, embedding data on children into global frameworks and commitments
- **Emergency Preparedness:** To provide preparedness planning estimates for individual and multiple hazards at sub-national level
- **Children climate risk analysis:** To conduct climate risk analysis at sub-national, national regional and global levels including global CCRI and sub-national CCRI-DRM
- **Climate Financing:** To support applications for climate financing such as GCF

Access: The database will provide improved and easy accessibility for country and regional offices, host governments and all stakeholders to support informed decision-making in diverse data formats (excel, geospatial, GeoAI chatbot).

1. [Global Child Hazard Database Visualization Dashboard](#) (access upon request to internal UNICEF until testing and review phase is completed)
2. [UNICEF Data warehouse](#) (planned for Q1 2026)
3. [Emergency Preparedness Platform](#) (internal UNICEF in 2026)
4. Geosphere, a GeoAI conversation tool to directly use natural language querying to understand, visualize and analyze the data (access restricted to the beta platform and available to UNICEF staff upon request)

The development of the Global Child Hazard Database is an ongoing process and will be continuously updated as and when new data sets are available.

Contact: For access/questions/comments, please contact the Climate and Environment Data unit

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